C3.4 Why is crude oil important as a source of new materials?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Crude oil is mixture of hydrocarbons. It is used as a source of fuels and as a feedstock for making chemicals (including polymers) for a very wide range of consumer products. Almost all of the consumer products we use involve the use of crude oil in their manufacture or transport. Crude oil is finite. If we continue to burn it at our present rate it will run out in the near future. Crude oil makes a significant positive difference to our lives, but our current use of crude oil is not sustainable. Decisions about the use of crude oil must balance short-term benefits with the need to conserve this resource for the future. Crude oil is a mixture. It needs to be separated into groups of molecules of similar size called fractions. This is done by fractional distillation. Fractional distillation depends on the different boiling points of the hydrocarbons, which in turn is related to the size of the molecules and the intermolecular forces between them. The fractions are mixtures, mainly of alkanes, with a narrow range of boiling points. The first four alkanes show typical properties of a homologous series: each subsequent member increases in size by CH_2, they have a general formula and show trends in their physical and chemical properties.	1. recall that crude oil is a main source of hydrocarbons and is a feedstock for the petrochemical industry 2. explain how modern life is crucially dependent upon hydrocarbons and recognise that crude oil is a finite resource 3. describe and explain the separation of crude oil by fractional distillation *PAG*3 4. describe the fractions of crude oil as largely a mixture of compounds of formula C_nH_{2n+2} which are members of the alkane homologous series 5. use ideas about energy transfers and the relative strength of chemical bonds and intermolecular forces to explain the different temperatures at which changes of state occur 6. deduce the empirical formula of a compound from the relative numbers of atoms present or from a model or diagram and vice versa 7. use arithmetic computation and ratio when determining empirical formulae M1c

Linked learning opportunities

Ideas about Science:

- decision making in the context of the use of crude oil for fuels and as a feedstock (IaS3)

C3.4 Why is crude oil important as a source of new materials?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
The molecular formula of an alkane shows the number of atoms present in each molecule. These formulae can be simplified to show the simplest ratio of carbon to hydrogen atoms. This type of formula is an empirical formula (IaS4). Small molecules like alkanes and many of those met in chapter C1 contain non-metal atoms which are bonded to each other by covalent bonds. A covalent bond is a strong bond between two atoms that formed from a shared pair of electrons. A covalent bond can be represented by a dot and cross diagram. Molecules can be shown as molecular or empirical formulae, displayed formulae (which show all of the bonds in the molecule) or in a 3 dimensional 'balls and stick' model.	8. describe the arrangement of chemical bonds in simple molecules 9. explain covalent bonding in terms of the sharing of electrons 10. construct dot and cross diagrams for simple covalent substances 11. represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures for simple molecules M5b 12. describe the limitations of dot and cross diagrams, ball and stick models and two and three dimensional representations when used to represent simple molecules	**Ideas about Science:** • the use and limitations of models to represent bonding in simple molecules (IaS3)
Simple molecules have strong covalent bonds joining the atoms within the molecule, but they only have weak intermolecular forces. No covalent bonds are broken when simple molecules boil. The molecules move apart when given enough energy to overcome the intermolecular forces. This explains their low melting and boiling points.	13. translate information between diagrammatic and numerical forms M4a 14. explain how the bulk properties of simple molecules are related to the covalent bonds they contain and their bond strengths in relation to intermolecular forces	
Cracking long chain alkanes makes smaller more useful molecules that are in great demand as fuels (for example petrol). Cracking also yields alkenes – hydrocarbons with carbon-carbon double bonds. Alkenes are much more reactive than alkanes and can react to make a very wide range of products including polymers. Without cracking, we would need to extract a lot more crude oil to meet demand for petrol	15. describe the production of materials that are more useful by cracking	**Ideas about Science**: • cracking as a positive application of science, to reduce extraction of crude oil and so conserves oil reserves

C3.4 Why is crude oil important as a source of new materials?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Alkanes and alkenes burn in plenty of air to make carbon dioxide and water. The double bond makes alkenes more reactive than alkanes. Addition across the double bond means that alkenes decolourise bromine water and can form polymers. An alcohol has a structure like an alkane, but with one hydrogen replaced by an OH group. Alcohols burn to make carbon dioxide and water, and can also be oxidised to make carboxylic acids. All of these compounds are useful to make consumer products. They have different properties due to their different functional groups. Alkanes do not have a functional group and so are unreactive. The functional group of alkenes – the double bond – is used for addition reactions. The OH functional group in alcohols give them a range of uses including their use as solvents that are miscible with water. The carboxylic acid functional group behaves as a weak acid, and these acids are found in foods and personal care products.	16. recognise functional groups and identify members of the same homologous series *(separate science only)* 17. name and draw the structural formulae, using fully displayed formulae, of the first four members of the straight chain alkanes and alkenes, alcohols and carboxylic acids *(separate science only)* 18. predict the formulae and structures of products of reactions (combustion, addition across a double bond and oxidation of alcohols to carboxylic acids) of the first four and other given members of these homologous series *(separate science only)* 19. recall that it is the generality of reactions of functional groups that determine the reactions of organic compounds *(separate science only)*	**Practical work:** • investigate the reactions of alkanes, alkenes and alcohols **Ideas about Science:** • the use of models to represent functional groups in homologous series (IaS3)